OSF Amendment 4 – GSSM Evaluation Protocol for H1 (P1)

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0. Registration status

This amendment makes the registered GSSM evaluation protocol (P1), which hypothesis H1 uses, operational and corrects it to match the GSSM paper and code. The hypotheses, tests and Holm family of the original registration are unchanged.

- **Original registration:** OSF Registries, 4 October 2026, https://osf.io/y5zrp. Earlier amendments: https://osf.io/cfdkj (1), https://osf.io/zd8jx (2), https://osf.io/hbfm9 (3).
- **Data access at filing:** the author has not accessed any version of the SHRP2 bird's-eye trajectory dataset, neither the complete version (DOI 10.15787/VTT1/EFYEJR) nor the public version (DOI 10.15787/VTT1/T7UUC1). No SHRP2 count, summary or result has informed this amendment. The published GSSM voting file is not opened before the data licence is granted.
- **Sources used:** the GSSM paper (Jiao et al., 2026; arXiv:2505.13556v5, Sections 5.2–5.3 and 6) and the authors' public code (github.com/Yiru-Jiao/GSSM), in particular `analyse_events.py`, `utils_evaluation.py` and `utils_eval_metrics.py`.

1. Reason for the amendment

The registration describes P1 as: "Frame-level scores in danger periods are compared with safe periods, as in Jiao et al. (2026): the danger period starts at the event start or 4.5 s before the event time, whichever is later." While building the code that runs GSSM on our splits, a full reading of the GSSM paper and code showed three differences. Each would have made the H1 comparison unlike the published protocol it claims to follow.

1. **Danger-period start.** The paper defines the danger period "from the start time, or 4.5 seconds before impact time if the annotated start time is less than 4.5 seconds until impact", and the code sets the start to min(impact − 4.5 s, start). The start is therefore the EARLIER of the two, not the later. The registration also omitted the end of the danger period and the definition of safe periods.
2. **Unit of counting.** GSSM does not compare frames. It counts per event and per surrounding object. A true positive is an alert lasting at least 0.5 s for the event's conflicting object in the danger period. False and true negatives are counted per other

object in its own safe period. The proposed model, LightGBM and the classical measures give one score per subject-vehicle frame, so a rule is needed for how their scores enter this object-level protocol.

3. **Voted subset.** GSSM chooses the conflicting object of each event by a vote among methods, and its code requires at least 15 voting methods. Reproducing that vote would need the ArgoverseHV dataset and about 11 further GSSM trainings, which the registration did not plan. The authors publish the result of their vote.

The corrections below make P1 operational. They follow the published protocol wherever it can be followed, and they were fixed before any SHRP2 data were accessed.

2. Corrected protocol P1

P1 is GSSM's evaluation exactly as coded by its authors. The functions named below are applied unchanged to every method. Times come from the event metadata (start, impact and end timestamps).

Element	Rule (GSSM code)
Danger period	from min(impact − 4.5 s, start) to min(impact + 0.5 s, end)
Conflicting object of an event (per method, first stage)	the object with the highest mean risk in the danger period; it needs at least 20 frames (2 s) there and data before the danger period, and its 25th, 50th and 75th risk percentiles must all be lower before the danger period than within it; otherwise the event has no recorded danger for that method (`determine_target`, `is_target_recorded`)
True positive / false negative	per event: more than 5 frames above the threshold for the conflicting object in the danger period (TP), otherwise FN
Safe period of another object	that object's frames before (start − 3 s), needing at least 35 of them; the window is its 16th to 65th frame (1.5 s after it was first recorded, about 5 s long); the subject vehicle never decelerates harder than 1.5 m/s² in the window

Element	Rule (GSSM code)
False positive / true negative	per object in a safe period: any frame above the threshold (FP), otherwise TN; counted only in events where the method records a danger period, as GSSM's code does (an event without one is skipped entirely)
AUPRC	precision and recall over the threshold grid, interpolated onto 1,000 recall points and integrated by the trapezoid rule (`get_auprc`)
mTTI	at the threshold with the highest F1 = TP / (TP + (FP + FN)/2): median over correctly detected events of impact time minus the last safe-to-unsafe switch at or before impact, restricted to TTI < 10 s (`issue_warning`, `get_time`)

Threshold grids (fixed now; they do not depend on any data):

- GSSM: its own grid of 100 values from −0.06 to 5.88 in steps of 0.06.
- TTC and 2D TTC (smaller = riskier): GSSM's TTC2D grid, round(u^7, 2) for u = 0, 0.0135, …, 1.93 (unique values).
- DRAC (larger = riskier): 0 to 10 m/s² in steps of 0.05.
- Proposed model and LightGBM (cumulative risk at 3 s, larger = riskier): the logistic function of t = −10, −9.9, …, +10 (201 values).

Best classical measure (H1b). As registered, it is chosen on validation data. Among TTC, DRAC, 2D TTC and 2D TTC with the moving-pair filter, the one with the highest P1 AUPRC on the validation events is chosen, before the test set is evaluated.

3. Methods that give one score per frame

The proposed model, LightGBM and the classical measures (TTC, DRAC and 2D TTC as registered: the subject vehicle's value at the frame, over all its objects) each give one score per subject-vehicle frame. In P1, that score is assigned to every surrounding object present in the test set at that frame (each row event × object × time of `event_data.h5`). GSSM's code then runs on these rows exactly as on its own scores.

- **GSSM** keeps its own pairwise score per object, computed by its own code (`evaluate_safety.py`).

- **Missing scores.** A frame where a method has no score (for the proposed model and LightGBM: less than 2 s of history, or a masked subject frame) counts as below every threshold. Such frames never issue an alert.
- **Ties.** Objects present at the same frames share a frame-level method's score. Where two objects tie for the highest mean risk in the danger period, GSSM's code picks the first in its own order (`idxmax`); this is kept.
- **What this means for interpretation.** A frame-level method cannot say which object is dangerous; it can only say when the subject vehicle is at risk. Under P1 it is credited when its alarm covers the conflicting object's danger period, and penalised when it alarms during other objects' safe periods. This limitation is stated whenever P1 results are reported.

4. Event sets

The primary H1 analysis uses all included test events with GSSM's first-stage rules. The two registered subsets use its third-stage rules, with one conflicting object shared by all methods.

Set	Events	Conflicting object	GSSM code
Primary (H1a, H1b)	all included events of the test split (registered inclusion = GSSM's: at least 5 s of object data, conflict not "none")	chosen by each method itself (first stage)	`parallel_records` , `issue_warning`
Sensitivity: GSSM's voted subset (registered)	test-split events with a conflicting object in the authors' published vote (`Conflicts/Voted_conflicting_targets.csv` , 4TU.ResearchData, doi 10.4121/9caa1e6c-9abd-4e36-ae28-c9ea4542d940)	the voted object	`evaluate`
Sensitivity: method-independent subset (registered)	test-split leading-vehicle conflicts whose nearest in-lane object ahead at impact time is recorded	that nearest object	`evaluate`

- In the primary set, each method's AUPRC uses the events where that method records a danger period, as in GSSM's first-stage evaluation. The number of such events is reported per method.
- The voted file is matched by event ID and object ID. An event whose voted object does not appear in the test set's object data is dropped and counted.

- The voted file is downloaded only after the data licence is granted. It holds results derived from SHRP2.

5. GSSM comparator

H1a compares the proposed model with **h-C**, the GSSM the paper reports as its headline model: trained on highD and using current features (Table 6 and Figure 5 of the paper). It learns from no SHRP2 data, so the registered rule that any SHRP2 data GSSM learns from come from the training split holds automatically.

- **Weights.** The authors' published trained h-C model (4TU.ResearchData, doi 10.4121/9caa1e6c-9abd-4e36-ae28-c9ea4542d940, `PreparedData`) is used. Only its highD model folder is extracted; the archive's other contents, which include features derived from SHRP2, are not opened.
- **Fallback.** If the published weights cannot be obtained or loaded, h-C is retrained with the authors' code and settings on highD (seed 131, as in their code). This is reported.
- **Secondary model (descriptive, not tested).** S-C is retrained with the authors' code and settings on the safe baseline trips of our training split only, with the code's own 80/20 train/validation division. The authors' segmentation code reads environment categories for baseline trips; where the data do not provide them they are entered as "Unknown". This does not affect the current features S-C uses.
- **Scores.** GSSM's own `evaluate_safety.py` scores the test events in the authors' environment (Python 3.10, their pinned packages), unchanged.

6. Inference frames, bootstrap and not-testable rules

Frames scored for P1. The danger period runs to 0.5 s after impact, and safe periods lie early in the record. P1 therefore needs scores at frames that protocol P2 leaves out of its labels. For P1 only, the registered models (all five seeds) and LightGBM score every subject frame of the included test events that has 2 s of contiguous history and an unmasked subject frame. This includes frames after the impact time and in the 5–7 s buffer. No model is retrained for this, and P2 labels and analyses are unchanged.

Uncertainty and tests. The registered paired cluster bootstrap over events is used (B = 2,000, seed 131).

1. Each resample draws test events with replacement. An event carries, for every method and threshold, its danger-period outcome (TP/FN or none) and its safe-object counts (FP, TN).
2. Per method, the counts are summed per threshold and the AUPRC is recomputed with `get_auprc`.

3. Δ* = proposed-model AUPRC averaged over the five seeds, minus the comparator's AUPRC. p = (1 + #{Δ* ⩽ 0}) / (B + 1), one-sided, as registered.

H1a uses GSSM h-C as the comparator and H1b the best classical measure. Both stay in the registered nine-test Holm family. mTTI is reported with GSSM's exact sign-test interval at 99%, as the paper states (the code's helper defaults to 95%), and is not tested.

Not testable. A comparison is not testable if fewer than 50 test events have a recorded danger period for either method in it, which is the registered small-domain threshold. Resamples with an undefined AUPRC (no true positive at any threshold) are discarded and counted. More than 5% discarded flags the test as unreliable, but it is still reported (registered rule).

7. What stays unchanged

Everything not named above remains as registered and amended:

- H1–H4 and their nine tests, the Holm family, the minimum effects of interest and protocol P2;
- the H5 family (Amendments 2 and 3);
- inclusion rules, labels, splits, models, seeds, budgets and sensitivity analyses.

The registered H1 statement "model ranks risk better than GSSM and the classical measures" and its primary endpoint (AUPRC under P1) keep their meaning. Only P1's operational definition changes.

Process note. The discrepancy was found on 11 October 2026 while writing the code that runs GSSM on our events and splits, before any SHRP2 access. The code implementing this amendment is tested on synthetic data in the SHRP2 format and committed to the project repository before the data are accessed.

Use of AI assistance. As stated in the original registration, code development, literature verification and drafting were assisted by an AI system; all analyses are run, checked and approved by the author.

Appendix A. Implementation clarifications for model training

The registration leaves the points below open. They were fixed in the training and analysis code and tested on synthetic data between 11 and 14 October 2026, before any SHRP2 data were accessed. They change no hypothesis, endpoint or test. They are filed here so that they carry a public date before access.

#	Registered text	Clarification
C1	Pretraining: best epoch by validation ADE (Section 4)	The validation part of the pretraining data is the public validation recordings plus the SHRP2 validation-split safe baselines, with the ADE pooled over these samples. No crash or near-crash event enters pretraining.
C2	Public pretraining data (Section 2)	INTERACTION keeps its development split. highD recordings are split 80/10/10 within each location by the project-wide rule (seed 131). Public test recordings are never used.
C3	H2 pretraining excludes every SHRP2 trip of the held-out level (Section 2)	Only trips known to be in the held-out level are left out. Trips with an unknown level stay (expected: all safe baselines, which carry no environment data). Models with identical pretraining sets share one encoder per seed; training is deterministic, so this equals retraining.
C4	Unknown categories are excluded from that factor's analyses (Section 3)	H2 models are fine-tuned and selected on trips with a known level only. If baselines have no environment data, H2 models use only the event trips of their level; negatives then come from the earlier frames of those trips.
C5	Size-matched model pretrained and fine-tuned on the subset only (Amendment 1)	No crash or near-crash may enter pretraining, so the subset replaces the larger level's trips in pretraining. The SHRP2 pretraining trips are the baselines not in the held-out level, and for the larger level only those in the subset. Subset order: a seeded permutation (the model's seed) of the sorted trip IDs.

#	Registered text	Clarification
C6	Alternative risk head and its selection rule (Sections 4 and 6)	The alternative head is trained with per-horizon binary cross-entropy on horizons whose outcome is known, with the best epoch chosen by that validation loss. The isotonic projection is the per-sample least-squares non-decreasing fit. The rule (AUPRC +0.01, adaptive ECE no more than 20% worse) is applied to five-seed means.
C7	Risk-head selection on validation data (Section 6)	The head is chosen after the central models of both heads (augmentation on, five seeds) and before any H2 model is trained. All later models use the chosen head.
C8	Training divergence rule (Section 6)	The rule is applied per training stage (pretraining or fine-tuning). A replaced model takes the next replacement seed (136, 137, …) not yet used by the same model type. Every event is logged.
C9	P1 scores (this amendment)	P1 frame scores come from the H1 models (central model, chosen head, augmentation on, five seeds) and from LightGBM: the cumulative risk at 3 s on every valid frame of the validation and test events.
C10	H3 federated models (Sections 4 and 6)	Neighbour augmentation is ON (the registered default) and the risk head is the one chosen by the registered selection. The development H3 runs had no augmentation. The best round is chosen by that head's pooled validation loss.
C11	H3 clients (Section 3)	Trips with an unknown level of lighting, surface or traffic density belong to no client. If safe baselines carry no environment data, H3 uses event trips only.

#	Registered text	Clarification
C12	Client merging: traffic density first, then lighting (Section 3)	Two steps. (1) Every (lighting, surface) pair with a cell below 100 training events becomes one client. (2) Every surface level with a client below 100 becomes one client. Training events are crash and near-crash trips of the training split. If a client is still below 100, or fewer than 2 clients remain, H3 is not testable.
C13	Divergence rule for H3 (Section 6)	A regime run with a non-finite loss is re-run once at half the learning rates, then with the next replacement seed not yet used by that regime. Every event is logged.
C14	S-C retrained on the training-split safe baselines (this amendment, Section 5)	S-C is segmented and trained in a separate workspace that holds training-split trips only. The authors' segmentation gives each surrounding object a random size drawn from the event targets in that workspace's metadata, so these sizes come from training-split events only. The authors' own 80/20 train/validation division is applied inside the training-split baselines. The trained model is copied unchanged into the test workspace to be scored.
C15	H2 small-domain rule: fewer than 50 test events at 3 s (Section 6)	A factor-direction pair is testable when the held-out level's test split holds at least 50 crash or near-crash trips with at least one positive frame at the primary horizon. Frames are not counted as events.
C16	Sensitivity 1, crash-only outcome (Section 7)	Near-crash trips are left out of evaluation (validation and test). Crash trips give the positives; baseline trips and earlier frames of crash trips give the negatives. Models are not retrained.

#	Registered text	Clarification
C17	H2 secondary criterion with five seeds (Amendment 1)	Per pair and α, the false-alarm rate is averaged over the five seeds. A violation is a 95% cluster-bootstrap lower bound of that seed mean above α. Clopper–Pearson bounds are reported per seed.
C18	Sensitivity 4, both risk heads (Section 7)	The head not chosen exists only for the central models, so sensitivity 4 is reported for the H1 key secondary endpoint and for H4 where both models exist.
C19	Sensitivity 6, temperature and isotonic recalibration (Section 7)	Each model is recalibrated on its own selection data (its domain level, its client, or all clients for FedAvg). For the separate head, one temperature minimises the binary cross-entropy on the registered grid before the per-sample isotonic projection; isotonic is per horizon, then a running maximum. Applied to H2 and H3a.
C20	Sensitivity 7, baselines with at least one detected object (Section 7)	A baseline trip is kept if any radar object is valid in any frame of any of its samples' 2-s windows; event trips are always kept.
C21	Method-independent subset (Section 7; this amendment, Section 4)	Included leading-vehicle conflicts. The conflicting object is the nearest object ahead within 1.75 m of the subject's heading line at the impact frame; events with no such object are left out; GSSM's third stage is applied with that object.
C22	H1 key secondary endpoint against GSSM under P2 (Section 1)	GSSM scores exist only for events, so this comparison uses the test samples of event trips. GSSM's frame score is its highest intensity over the objects at that frame; frames with no object score below every GSSM score.

#	Registered text	Clarification
C23	Sensitivity 8, no handling of implausible values (Section 7)	Full repetition: the whole pipeline is repeated on the data loaded without masking. Samples, classical measures and features are rebuilt, every model is retrained with the same seeds, budget and rules, the registered risk-head rule is re-applied (the choice is reported), and every confirmatory test is rerun. The split stays fixed, so trips excluded as implausible stay out. GSSM is unaffected, since its inputs were never masked. The two runs are kept fully apart.

References

Jiao, Y., Calvert, S. C., van Cranenburgh, S., & van Lint, H. (2026). Learning collision risk proactively from naturalistic driving data at scale. *Nature Machine Intelligence*. https://doi.org/10.1038/s42256-026-01189-w (preprint arXiv:2505.13556v5). Code: https://github.com/Yiru-Jiao/GSSM.

Jiao, Y., & Calvert, S. C. (2025). *Bird's eye view trajectory reconstruction of naturalistic crashes and near-crashes in the SHRP2 NDS* [Data set]. VTTI Dataverse. https://doi.org/10.15787/VTT1/EFYEJR